



The effects of the number of channels and gyroscopic data on the classification performance in EMG data acquired by Myo armband

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ABSTRACT

Processing and classification of Electromyography (EMG) signals is a common practice in prosthetic arm design for hand amputees. This study investigates how reliable classification results may be obtained from less muscle data, as is the case for the muscle loss in hand amputees. In order to increase classification performance, features from gyroscopic data, not previously used in studies in the literature, were investigated. Data was acquired from 10 normal subjects using the Myo armband for 7 hand gestures: fist, fingers spread, wave-in, wave-out, pronation, supination, and rest. Subjects repeated each gesture 30 times. EMG signals were preprocessed to extract features. Twenty (20) features were used in the feature matrix; 14 time domain and 6 frequency domain. Features were selected to determine the highest accuracy using the Support Vector Machine (SVM) and k-Nearest Neighbor (KNN) as classification algorithm. The Classification Learner application in Matlab® was used for classification. The highest accuracy using all EMG channels was 98.38 %. When the number of channels was reduced to 3, the accuracy was over 90 %. It is observed that gyroscopic features increase the performance when a small number of EMG channels is used.

1. Introduction

Biological signals that occur as a result of muscle contraction are called Electromyography (EMG). Voluntary muscle movements occur when electrical signals reach the muscles from the brain via nerves. Just as the contraction of the muscle fibers is stimulated by electrical conduction of the nerves, the contraction of the muscle fibers creates electrical signals; the EMG signals [1]. Two main types of electrodes are used to detect EMG signals: needle and surface [2]. Surface electrodes are used in the control of artificial limbs because they are more comfortable and do not require surgical intervention [3,4]. In this study, the Myo armband with 8 dry surface electrodes was used for data acquisition. Dry electrodes on the Myo armband allow easy use of the armband without preparation [5].

Controlling a prosthetic arm using the Myo armband and EMG signals is a common practice in the literature [2,6,7]. Fig. 1 shows the typical stages of processing of the EMG data for feature extraction and classification. High-pass filters of different cut-off frequencies including 5 Hz [8,9], 10 Hz [10] and 15 Hz [11] have been used as part of the preprocessing of EMG data. Features including root mean square (RMS) [12], waveform length (WL) [13], EMG spectrograms [9], short time

energy and zero crossing rate [14] are used to determine the region of muscle activation in the EMG signal. Sliding windows with length 100 ms [14], 240 ms [2], 250 ms [11,15,16], 260 ms [5] are used for data segmentation. Root mean square (RMS), mean absolute value (MAV), zero crossing (ZC), waveform length (WL) and slope sign change (SSC) features are frequently used in the feature matrix [2,9,11,12,14,15,17–19]. Principal Component Analysis (PCA) has been used in some studies to reduce the dimensionality of the feature matrix [15,18]. Many methods have been used for the classification of EMG data in the literature including: Random Forest [12], Hidden Markov Model-Gaussian Mixture Model (HMM-GMM), Convolutional Neural Networks (CNN) [5], Convolutional Network (ConvNet) [17], Dynamic Time Warping (DTW) [8], Artificial Neural Networks (ANN) [6], Multi-Layer Perceptron Neural Network (MLPNN) [20], k-Nearest Neighbor (KNN) [8], Linear Discriminant Analysis (LDA) [11] and Support Vector Machine (SVM) [2,7,9]. In addition, normalization of the EMG data [12], CNN architecture developed with transfer learning techniques [5], and presentation of a transfer learning scheme using ConvNet [17], have been proposed for each method to increase classification performance.

The effect of sampling frequency on classification performance has been examined and it was seen that high accuracy could be obtained by

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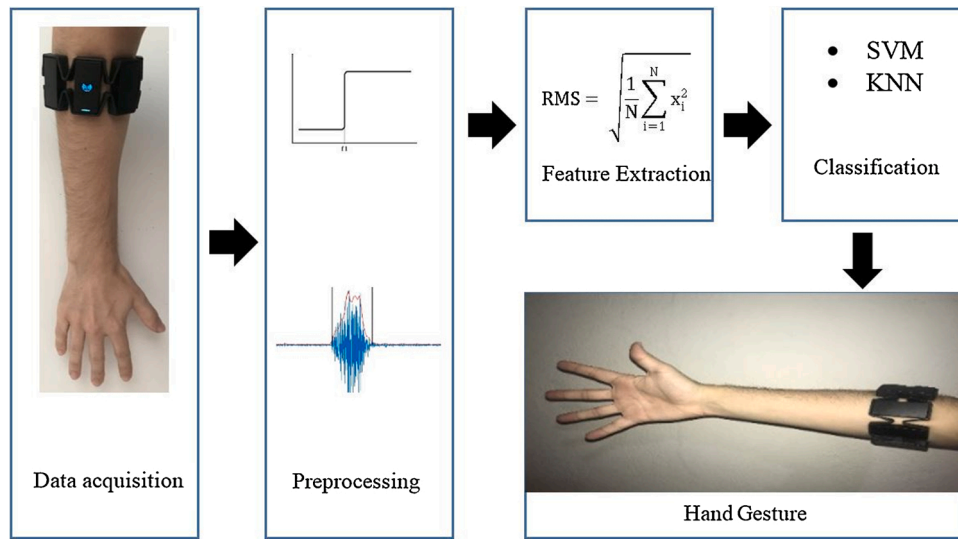


Fig. 1. Flow diagram for EMG Processing.

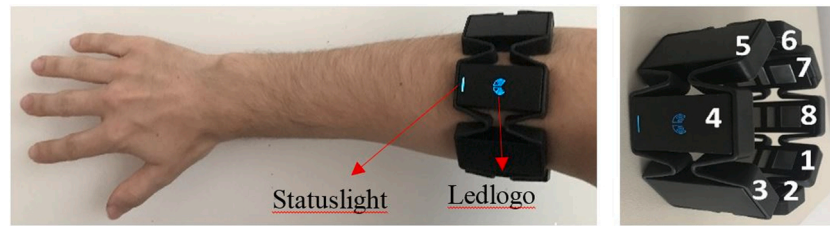


Fig. 2. Myo armband and its placement on the right forearm.

selecting feature sets at low sampling frequencies [16]. In addition, a study conducted on the effect of the amount of limb loss for hand amputees on classification performance determined that performance decreased with increasing limb loss [21].

The aim of this study is to obtain reliable classification of hand gestures using the least possible number of EMG channels, thereby simplifying the use of prosthetic arms for hand amputees, as the forearm muscles are both weak and their number is limited, which makes it difficult to acquire multi-channel EMG data. This study investigates the effect of reducing the number of channels in the Myo armband on classification performance. The study incorporates gyroscopic data and features to determine if inclusion improves classification performance, an aspect not previously reported in the literature.

2. Material and method

2.1. Data acquisition

2.1.1. Myo armband

The Myo armband is a wireless wearable technology device designed by Thalmic Labs in 2014 [2]. The Myo armband is based on surface EMG and IMU data. It weighs 93 g and has an adjustable diameter between 19–34 cm [21]. It has 8 EMG sensors and a 9 axis IMU sensor including 3 axis gyroscope, 3 axis accelerometer, and 3 axis magnetometer [22]. EMG data is acquired at 200 Hz and IMU data at 50 Hz sampling frequency [23]. However the Myo armband samples the EMG data at a lower frequency than preferred due to embedded system constraints

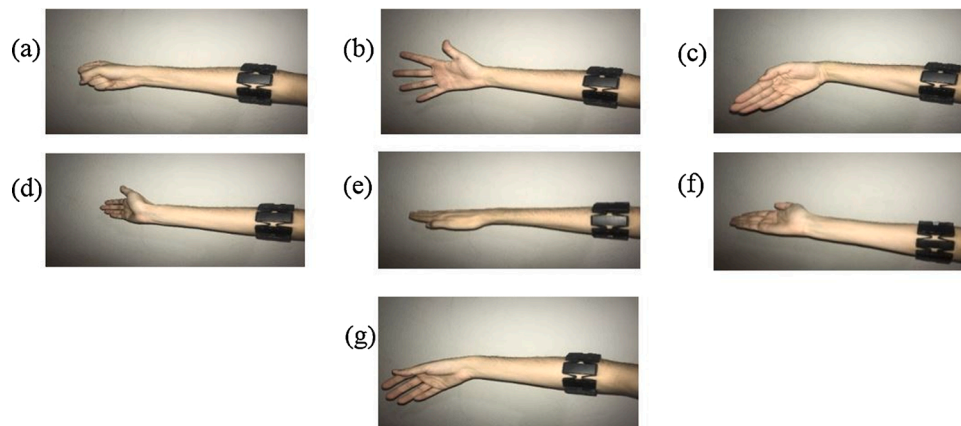


Fig. 3. Classified hand gestures (a) fist (b) finger spread (c) wave-in (d) wave-out (e) pronation (f) supination (g) rest.

Table 1
Time domain features.

Abbreviation = Feature = Equation	Abbreviation = Feature = Equation
$F1 = \text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$	$F2 = \text{MAV} = \frac{1}{N} \sum_{i=1}^N x_i $
$F3 = \text{ZC} = \sum_{i=1}^{N-1} [\text{sgn}(x_i * x_{i+1}) \cap x_i - x_{i+1}] \geq \text{threshold}$	$F4 = \text{SSC} = \sum_{i=2}^{N-1} [f((x_i - x_{i-1}) * (x_i - x_{i+1}))]$
$\text{sign}(x) = \begin{cases} 1, & x \geq \text{threshold} \\ 0, & x < \text{threshold} \end{cases}$	$f(x) = \begin{cases} 1, & x \geq \text{threshold} \\ 0, & x < \text{threshold} \end{cases}$
$F5 = \text{WL} = \sum_{i=1}^{N-1} x_{i+1} - x_i $	$F6 = \text{VAR} = \frac{1}{N-1} \sum_{i=1}^N x_i^2$
$F7 = \text{WAMP} = \sum_{i=1}^{N-1} [f(x_i - x_{i+1})]$	$F8 = \text{MYOP} = \frac{1}{N} \sum_{i=1}^N f(x_i)$
$f(x) = \begin{cases} 1, & x \geq \text{threshold} \\ 0, & x < \text{threshold} \end{cases}$	$f(x) = \begin{cases} 1, & x \geq \text{threshold} \\ 0, & x < \text{threshold} \end{cases}$
$F9 = \text{AAC} = \frac{1}{N} \sum_{i=1}^{N-1} x_{i+1} - x_i $	$F10 = \text{SSI} = \sum_{i=1}^N x_i^2$
$F11 = \text{DASDV} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (x_{i+1} - x_i)^2}$	$F12 = \text{IEMG} = \sum_{i=1}^N x_i $
$F13 = \text{MMAV1} = \frac{1}{N} \sum_{i=1}^N w_i * x_i $	$F14 = \text{MMAV2} = \frac{1}{N} \sum_{i=1}^N w_i * x_i $
$w_i = \begin{cases} 1, & 0.25N \leq i \leq 0.75N \\ 0.5, & \text{others} \end{cases}$	$w_i = \begin{cases} 1, & 0.25N \leq i \leq 0.75N \\ \frac{4i}{N}, & i < 0.25N \\ \frac{4(i-N)}{N}, & \text{others} \end{cases}$

[16]. Fig. 2 shows the placement of the Myo armband on the right forearm and the channel numbers.

2.1.2. Experimental procedure

EMG and gyroscopic data were acquired from 3 male and 7 female normal subjects (non-amputee) aged 33 ± 16 years. The subjects wore the Myo armband on their right forearm, close to the elbow, and held their arm straight out, perpendicular to the ground, as shown in Fig. 3. Subjects were asked to repeat six (6) hand gestures 30 times, the gestures being: fist, fingers spread, wave-in, wave-out, pronation and supination. Each gesture was recorded for 5 s, with the subject starting to move their hand from the rest gesture in the 2nd second and returning to rest at will and before the end of the recording. The data in the first two seconds before moving to the gesture were considered as the rest gesture; classification was made for the 7 gestures shown in Fig. 3. The data set and user manual used in this study can be downloaded from Appendix A.

2.2. Data segmentation

The EMG signal is inherently noisy; therefore, a data window is applied to delineate the segment of interest in order to extract the descriptive features. In general, the longer the data window, then the greater the accuracy of the classification presented to the control system [24]. However, as the window length increases, so does the delay for a classification to be made. For this reason, the window length is

recommended to be constrained to less than 300 ms in a real time system [25], with a compromise between low delay and accuracy of classification.

A short window length decreases the accuracy of classification, as it increases the variance of the features [25,26]. It has been found that a window length less than 125 ms causes too high variance for accurate classification [25], and so a window length of 150 ms has been chosen in this study for performing the feature extraction. The system delay will be directly related to the window length. The total processing time for all the stages to process the EMG signal must be less than the window length as all preprocessing, feature extraction and classification must be complete before the next window, and must be less than the real time constraint of 300 ms [25].

Windowing can be performed in two ways, adjacent and overlapping. In adjacent windowing, each new window begins where the previous window ends. In overlapping windowing, a window will start before the previous window ends, with the windows overlapping by a given amount. An overlap for windowing of 50 % has been selected in many of the studies reviewed [2,14–16,27,28].

2.3. Preprocessing

2.3.1. EMG filtering

A raw EMG signal contains frequencies between 0–1000 Hz [29], with the significant frequencies between 0–400 Hz [30]. However, the

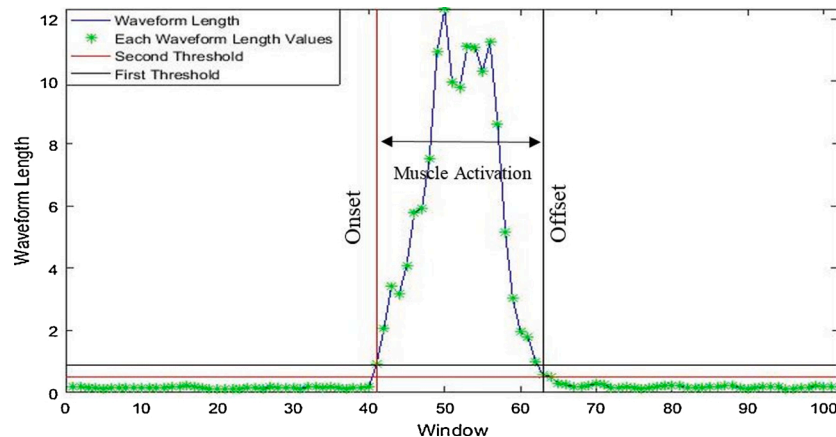


Fig. 4. Start and end of muscle activation region based on waveform length.

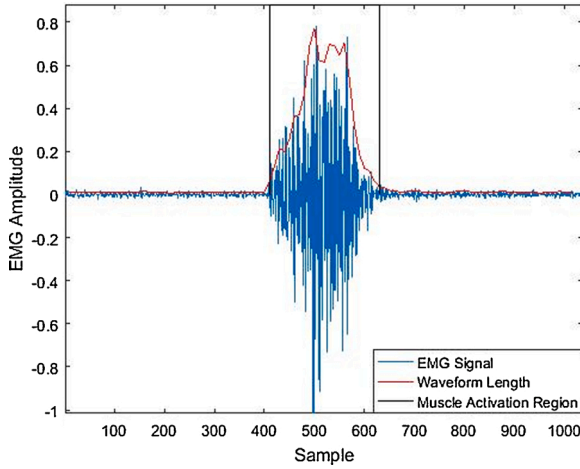


Fig. 5. Muscle activation region representation on an EMG signal.

Myo armband uses a sampling frequency of 200 Hz, which will limit the maximum frequency contained in the EMG to 100 Hz, as given by the Nyquist theorem.

A high-pass filter of 10 Hz has been applied to the EMG signal in order to remove body noise that interferes with the EMG signal [10]. The Myo armband includes a 50 Hz notch filter to remove power-line noise [31].

2.3.2. Finding the region of muscle activation in EMG signals

The region of muscle activation is found by dividing the EMG signal into windows of length 100 ms with 50 % overlap, with the waveform length (WL) being calculated for each window (see F5 in Table 1).

Threshold values to determine the onset and offset of muscle activation in each channel are based on the values of waveform length. The first 38 waveform length values (2 s) are used to represent the resting value. The onset threshold is chosen to be 2.2 times the largest value in these first 38 values. Start of motion is determined as the point when the waveform length value exceeds the onset threshold. A second, higher threshold is used to determine when the gesture ends, as it was observed that the muscles continued to contract after the gesture was finished. The second threshold is set at a value that is 1.7 times the first threshold. The end of the gesture is determined as the point when the waveform length value is below the offset threshold. The threshold values were determined empirically from the entire data set [13]. Our method to determine the muscle activation region was compared against other methods in the literature [9,12,14] and was found to give the best results.

Onset and offset of a gesture are found with respect to the window of waveform length. These points must be mapped as onset and offset in the EMG signal. For this study, the center point of the waveform length window is used as the sample point of the EMG signal. The muscle activation region detected by this method for a sample waveform is shown on the waveform length signal in Fig. 4 and on the EMG signal in Fig. 5.

2.3.3. Finding the region of muscle activation in gyroscopic signals

Methods used to find the muscle activation region in EMG signals have also been tried for gyroscope signals. However, the muscle activation region was not found correctly. This study has investigated a novel algorithm. For the method, the absolute value of the gyroscopic signal (x channel) is used. The maximum value from the first 90 points of the recording (representing rest) is used as a threshold. The signal is divided into adjacent windows of 200 ms length. Each window is checked until a window is found with a value that exceeds the threshold; the start of that window is taken as the start of the hand gesture. The gesture continues until the maximum value in a window falls below the

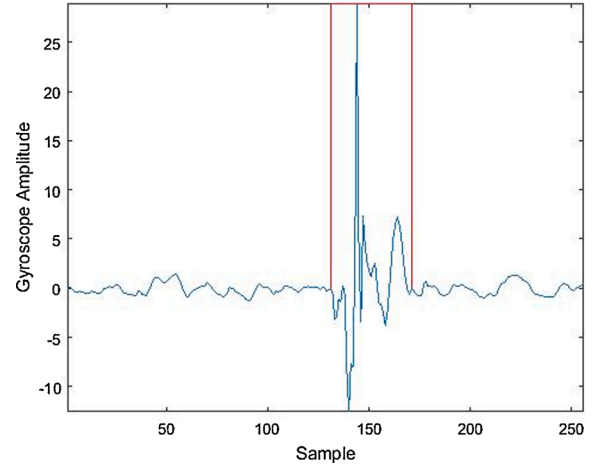


Fig. 6. Muscle activation region in a gyroscopic signal.

threshold value. An example of the muscle activation region detected by this method is shown in Fig. 6.

2.4. Feature extraction

In our approach, we determine 14 time domain features and 6 frequency domain features for each of the 150 ms signal windows for each EMG channel that we investigate for ability to classify gestures. We average the value of each feature from the windows that occur during muscle activation to form a scalar value for that feature. The process is repeated for each of the 8 channels of EMG data. Thus, each gesture is expressed by a 1×8 vector for each feature in the 8 channels. We create a data ensemble consisting of the 30 recordings of the 7 gestures for the 10 participants to form a matrix of dimension $2100 \times 8 \times 20$ for the 20 features in each of the 8 channels of EMG data.

The same equations and approach were used for feature extraction in the EMG and gyroscopic data. Numbered abbreviations are used to denote for which data type the features are calculated. For example, F1 refers to the RMS value of the EMG signal, and G1 refers to the RMS value of the gyroscopic signal.

2.4.1. Time domain features

Fourteen (14) time domain features were extracted in this study: Root Mean Square (RMS), Mean Absolute Value (MAV), Zero Crossing (ZC), Slope Sign Change (SSC), Waveform Length (WL), Variance (VAR), Willison Amplitude (WAMP), Myopulse Percentage Rate (MYOP), Average Amplitude Change (AAC), Simple Square Integral (SSI), Difference Absolute Standard Deviation Value (DASDV), Integrated EMG (IEMG), Modified Mean Absolute Value Type 1 (MMAV1), Modified Mean Absolute Value Type 2 (MMAV2) [32]. The time domain features are defined in Table 1. N represents window length, x_i represents the i -th EMG signal amplitude in a window.

2.4.2. Frequency domain features

Frequency domain features of the EMG signal were determined from the power spectral density for each of the time domain windows in each

Table 2
Frequency domain features.

Abbreviation = Feature = Equation	Abbreviation = Feature = Equation
$F15 = PKF = \max(P_i)$	$F16 = MNP = \frac{\sum_{i=1}^N P_i}{N}$
$F17 = TTP = \sum_{i=1}^N P_i$	$F18 = MNF = \frac{\sum_{i=1}^N P_i f_i}{\sum_{i=1}^N P_i}$
$F19 = SM1 = \sum_{i=1}^N P_i f_i$	$F20 = SM2 = \sum_{i=1}^N P_i f_i^2$

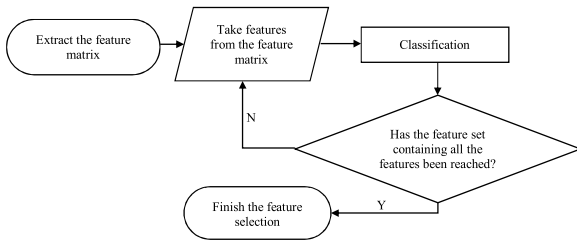


Fig. 7. Sequential forward selection flow diagram.

channel of a recording; a Hamming window function being applied to prevent spurious frequencies caused by a discontinuity at the edges of the period [33]. Six (6) frequency domain features were used in this study: Peak Frequency (PKF), Mean Power (MNP), Total Power (TTP), Mean Frequency (MNF), Spectral Moment 1 (SM1), Spectral Moment 2 (SM2) [32]. The frequency domain features are defined in Table 2. P_i represents the i -th power spectral density in a window, f_i represents the i -th frequency.

2.5. Classification

The accuracy of our classification is determined by dividing the data set into training and test sets. The division is pseudo-random, and is formed by placing recordings alternatively to the training set and to the test set. The division was made as 50 % training and 50 % test.

Twenty two (22) classification algorithms from the Classification Learner application in Matlab® have been tested to determine the optimum. SVM and KNN algorithms gave the greatest accuracy and have been used in this study. SVM is based on the principle of determining the boundaries of the data of classes that are separated from each other in the most appropriate way. In KNN, the k closest data to a class center are determined to form the boundary for that class. Test data are then determined within which class they lie. The k -fold cross validation method was used to evaluate the classification performance and $k = 5$ was chosen [34].

2.6. Feature selection

Extracting all features for the entire data set produces a high-dimensional feature matrix that increases processing time and decreases accuracy. In order to prevent this, the features with highest accuracy should be selected from among the feature matrix, forming a feature matrix with lower dimension and higher accuracy [32].

Sequential forward selection was used to select the features with highest performance from the feature matrix. In the first stage of this method, each feature was classified separately and the feature with the highest accuracy selected for the second stage. In the second stage, the accuracy of feature sets consisting of combinations of two features were considered by combining a further feature with the best feature selected from the first stage. The combination of features with the highest accuracy is then selected for the next stage. The process continues until all features have been combined to the feature set. In this way, optimum combinations of features are considered and the feature set with highest accuracy may be selected [35]. The flow diagram for these operations is shown in Fig. 7.

3. Results

3.1. Sequential forward selection results

Sequential forward selection was used to determine both the feature set and the classifier to be used for processing the EMG signals. In the first stage, the F11 (DASDV) feature gave the highest accuracy with 96.95 %. Sequential forward selection was continued until the feature

Table 3

Sequential forward selection third stage.

Features	Accuracies (%)					
	Q- SVM	C- SVM	M-SVM	F- KNN	W-KNN	S-KNN
F11+F7+F1	97.43	97.81	96.95	96.67	95.43	80.67
F11+F7+F2	97.71	97.90	97.14	96.29	95.43	80.95
F11+F7+F3	95.90	95.52	95.62	91.90	93.52	71.71
F11+F7+F4	95.71	96.10	95.14	91.14	94.00	71.90
F11+F7+F5	97.24	97.90	97.14	96.10	95.43	83.24
F11+F7+F6	96.48	96.76	96.19	94.86	93.62	79.33
F11+F7+F8	97.62	97.62	97.71	93.33	96.29	81.24
F11+F7+F9	97.71	98.38	97.14	95.71	94.86	78.86
F11+F7+F10	96.95	97.14	96.19	94.67	94.19	81.24
F11+F7+F12	97.90	97.90	97.14	96.19	95.52	82.76
F11+F7+F13	97.71	97.62	97.05	96.48	95.62	81.33
F11+F7+F14	97.62	98.00	96.86	96.19	95.62	80.76
F11+F7+F15	96.67	96.86	96.57	94.84	93.90	79.24
F11+F7+F16	96.95	97.14	96.57	97.76	94.19	90.19
F11+F7+F17	96.67	91.33	96.48	94.76	94.19	79.81
F11+F7+F18	96.95	97.76	96.48	95.05	93.33	80.57
F11+F7+F19	93.33	97.24	96.57	94.95	93.24	78.29
F11+F7+F20	96.57	96.19	95.52	91.81	91.90	73.43

matrix containing all the features was reached. Highest accuracy was obtained in the third step and the results for the best six classifiers are shown in Table 3. The six classifiers considered are: Quadratic SVM (Q-SVM), Cubic SVM (C-SVM), Medium Gaussian SVM (M-SVM), Fine KNN (F-KNN), Weighted KNN (W-KNN) and Subspace KNN (S-KNN).

The highest accuracy obtained at each stage up to 20 features is shown in Fig. 8. Although local maxima occur in the seventh, eighth, fifteenth and sixteenth stages, the highest accuracy occurred in the third stage, and was obtained with the C-SVM classifier for the feature set of F11+F7+F9. Fig. 8 therefore shows the accuracies at each stage obtained with C-SVM.

The outcome from using the F11 + F7 + F9 (DASDV + WAMP + AAC) feature set and the C-SVM algorithm for classification is given in the confusion matrix in Table 4 and the performance measurements in Table 5.

3.2. Adding gyroscopic features

The effect of combining gyroscopic data on the accuracy of classification of EMG signals has been examined by adding gyroscopic data to the feature matrix. Gyroscopic features have been tested for two EMG feature sets: F11 and F11 + F7 + F9. Gyroscopic features were tested one by one to determine which feature gave best results; the outcomes are shown in Figs. 9 and 10. When using gyroscopic data with one EMG feature, the G10 (SSI) feature gave the highest accuracy of 98 % from the time domain features, and the G15 (PKF) feature gave the highest accuracy of 98 % from the frequency domain features. Using three EMG features, the G6 (VAR) feature gave the highest accuracy of 98.1 % from the time domain features, and the G20 (SM2) feature gave the highest accuracy of 98.19 % from the frequency domain features.

3.3. Channel reduction results

The classification accuracy obtained for fewer EMG channels was investigated. All combinations of channels of up to seven channels, starting with two channels, were compared. Channel reduction was done for F11, which gave the highest accuracy with a single feature, and the F11 + F7 + F9 feature set, which gives the highest accuracy. The accuracy obtained by adding gyroscopic x channel data for all channel combinations was also compared. The combinations of channels with the highest accuracy are shown in Figs. 11 and 12.

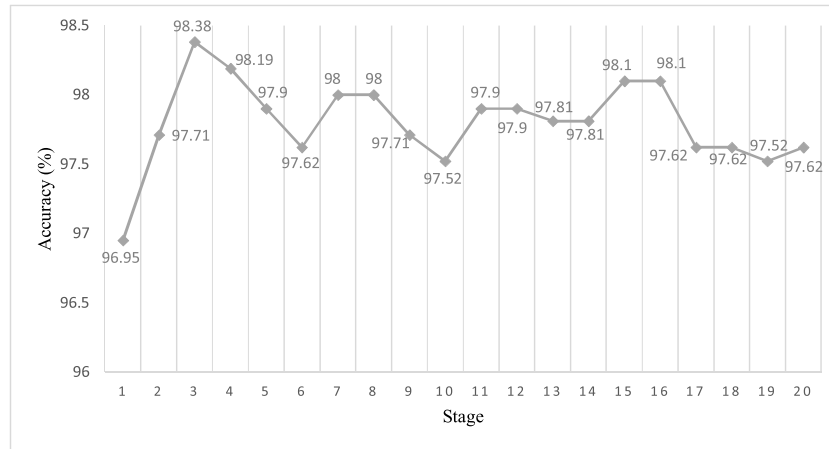


Fig. 8. Accuracy of C-SVM with successive stage.

Table 4
Confusion matrix.

		Predicted Class						
		Fist	Fingers Spread	Wave-in	Wave-out	Pronation	Supination	Rest
True Class	Fist	148	0	2	0	0	0	0
	Fingers Spread	0	150	0	0	0	0	0
	Wave-in	1	0	149	0	0	0	0
	Wave-out	0	0	0	149	0	1	0
	Pronation	0	0	0	1	138	11	0
	Supination	0	0	0	0	1	149	0
	Rest	0	0	0	0	0	0	150

Table 5
Performance measurements.

	Sensitivity (%)	Specificity (%)	Positive Predictive Value (%)	Negative Predictive Value (%)
Fist	98.67	99.89	99.33	99.78
Fingers Spread	100	100	100	100
Wave-in	99.33	99.78	98.68	99.89
Wave-out	99.33	99.89	99.33	99.89
Pronation	92	99.89	99.28	98.68
Supination	99.33	98.67	91.98	99.89
Rest	100	100	100	100
Average	98.38	99.73	98.37	99.73

4. Discussion

4.1. Feature selection and classification performance

Each feature, other than F3 (ZC) and F4 (SSC), had an accuracy of close to 90 % in the first stage of the sequential forward selection (Fig. 8). The poor performance of ZC and SSC may be due to the low sampling frequency (200 Hz) of the Myo armband. However, performance is compensated by the selection of other features [16].

When comparing performance to determine separate gestures (Table 5), finger spread and rest were able to be distinguished from other gestures without error. However, the ability to identify pronation was lower than other gestures and was frequently confused with supination (Table 4) [36]. This would be expected as forearm pronation and

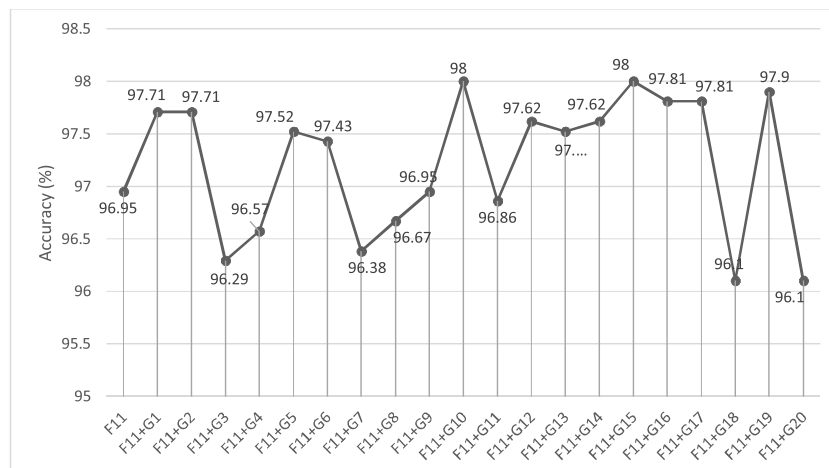


Fig. 9. C-SVM accuracies when using one EMG feature (F11) + Gyroscopic feature.

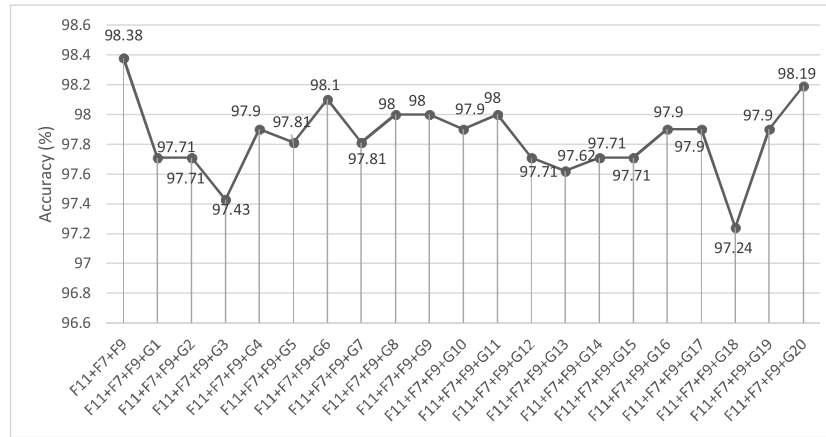


Fig. 10. C-SVM accuracy when using three EMG feature (F11+F7+F9) + Gyroscopic feature.

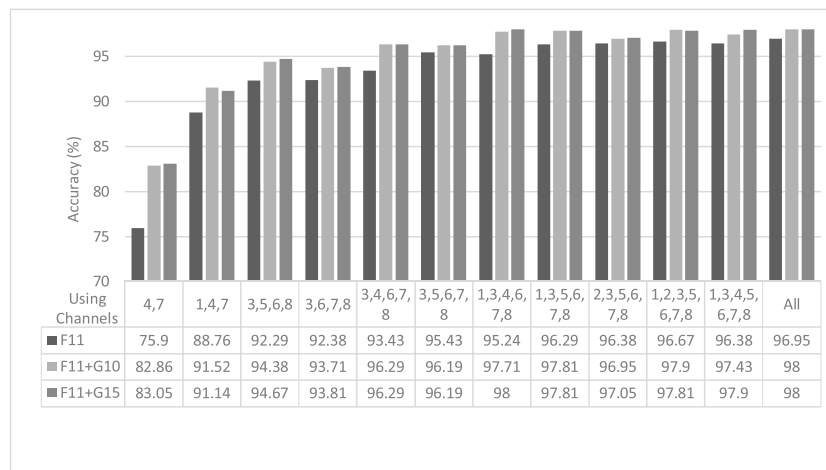


Fig. 11. Accuracy for number of channels: C-SVM accuracy using one EMG feature (F11) + Gyroscopic feature.

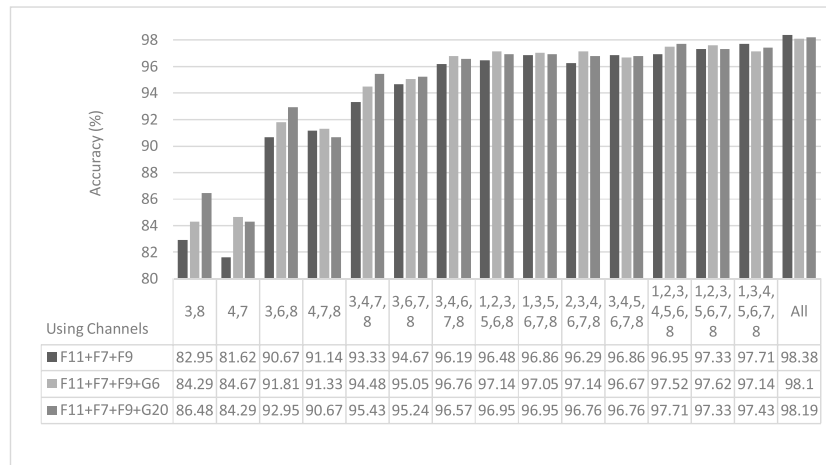


Fig. 12. Accuracy for number of channels: C-SVM accuracy using three EMG features (F11+F7+F9) + Gyroscopic feature.

supination both activate the two pronator and two supinator muscles close to the bones [11], which as a result makes it difficult to classify these gestures from surface EMG signals.

4.2. Classification with gyroscopic features

Fig. 9 shows how accuracy is increased when a gyroscopic feature is added to the F11 feature. The greatest accuracy of 98 % was obtained with G10, which adds 1.05 % to the accuracy. When adding a feature from the frequency domain, adding gyroscopic data gave 98 % accuracy

Table 6

Accuracy for each subject (%).

Subject	Q-SVM	C-SVM	M-SVM	F-KNN	W-KNN	S-KNN
1. Subject	99.05	99.05	98.1	96.19	84.76	88.57
2. Subject	98.1	96.19	100	96.19	95.24	82.86
3. Subject	96.19	95.24	97.14	96.19	95.24	83.81
4. Subject	100	100	100	97.14	97.14	86.67
5. Subject	100	100	100	100	100	83.81
6. Subject	100	100	98.1	97.14	94.29	78.1
7. Subject	99.05	99.05	100	98.1	96.19	86.67
8. Subject	100	100	100	100	99.05	91.43
9. Subject	100	99.05	99.05	100	99.05	88.57
10. Subject	100	100	100	100	100	87.62
Average	99.23	98.85	99.23	98.1	96.1	85.81
Entire Data Set	97.71	98.38	97.14	95.71	94.86	78.86

using G15, improving accuracy by 1.05 %.

Fig. 10 shows that adding gyroscopic data to F11 + F7 + F9 features did not improve accuracy. Adding G6 to the F11 + F7 + F9 features gave the best result, with an accuracy of 98.1 %, but this was 0.28 % lower than the EMG features alone. When adding a feature from the frequency domain, adding the G20 feature gave an accuracy of 98.19 %; 0.19 %

less than the EMG features alone. Adding gyroscopic data to three EMG features gave no benefit to accuracy.

4.3. Channel reduction

The effect of the number of channels on accuracy was investigated using the F11 feature and the F11 + F7 + F9 features. Fig. 11 shows the results for various combinations and numbers of channels using F11 and one gyroscopic feature. With 3–7 EMG channels, the accuracy ranges from 88.76 % to 96.67 % for F11; 91.52 % to 97.9 % for F11 + G10; and 91.14 % to 98 % for F11 + G15. Fig. 12 shows the results for various combinations and numbers of channels with F11 + F7 + F9 and one gyroscopic feature. With 3–7 EMG channels, the accuracy ranges from 90.67 % to 97.71 % for F11 + F7 + F9; 91.33 % to 97.62 % for F11 + F7 + F9 + G6; and 90.67 % to 97.71 % for F11 + F7 + F9 + G20. Our results provide reliable classification results for 3–7 channels [11,20,25]. Additionally, the gyroscope features increased the classification performance with reduced channels. Our results would indicate that a smaller number of appropriately placed EMG channels can be used in prosthetic arm control; in our case 3 channels in place of 8 channels. This has the further advantage of decreasing the dimension of the feature

Table 7

Comparison with studies using entire dataset.

Study	Number of EMG Channels	Number of Subjects	Number of Hand Gestures	Number of Each Hand Gesture Repeat	Features	Classifier	Highest Accuracy (%)
[12]	8	10	3	10–12	MAV, ZC, WL, SSC	Random Forest	96.40
[11]	8	5	22	4	RMS, MAV, ZC, WL, SSC, AR	LDA	95.00
[5]	8	First Subset: 18 Second Subset: 17	7	First Subset: 1 Second Subset: 3	CNN	CNN	97.81
[17]	8	First Subset: 19 Second Subset: 17	7	First Subset: 1 Second Subset: 3	CWT	ConvNet	98.31
[7]	8	23	4	10	MAV, STD, AR	SVM	89.93
[14]	8	1	6	100	RMS	GMM-HMM	99.00
[8]	8	10	5	10	Envelope	KNN-DTW	89.50
[9]	8	9	5	5	MAV, SSC, WL, RMS, Hjort	SVM	89.00
[6]	8	1	2	5	RMS	ANN	92.68
[20]	8	3	14	2	RMS, STD, MIN, MAX	MLPNN	90.05
[38]	8	15	5	X	MAV, RMS, VAR, STD	SVM	92.00
This Study	8	10	7	30	DASDV, WAMP, AAC	SVM	98.38
This Study	7	10	7	30	DASDV, Gyro SSI	SVM	97.90
This Study	6	10	7	30	DASDV, Gyro PKF	SVM	98.00
This Study	5	10	7	30	DASDV, WAMP, AAC, Gyro VAR	SVM	96.76
This Study	4	10	7	30	DASDV, WAMP, AAC, Gyro SM2	SVM	95.43
This Study	3	10	7	30	DASDV, WAMP, AAC, Gyro SM2	SVM	92.95

Table 8

Comparison with other studies using subject average.

Study	Number of EMG Channels	Number of Subjects	Number of Hand Gestures	Number of Each Hand Gesture	Features	Classifier	Average Accuracy (%)
[2]	8	6	7	8	RMS, MAV, SSC, WL, AR	SVM	95.26
[37]	8	8	9	5	WL, MAV, WAMP, CARD, SSC, ZC	LDA	91.67
[37]	6	8	9	5	WL, MAV, WAMP, CARD, SSC, ZC	LDA	90.14
[15]	8	9	6	X	MAV, ZC, SSC, WL, AR	SVM	99.02
This Study	8	10	7	30	DASDV, WAMP, AAC	SVM	99.23

matrix and thus the processing time.

4.4. Comparison with previous studies

Previous studies on EMG classification provide their results as either the average of the accuracies obtained from all subjects [2,15,37] or the accuracies obtained from the entire data set. This study compares its results with previous studies using each approach. The accuracy for each subject in this study and their average is shown in Table 6; a comparison with the studies in which the results obtained by using the entire data set is given in Table 7; and with studies in which the average of subject accuracy is given in Table 8. Table 7 also contains our results for reducing the number of channels. Note that one study in Table 7, although it had a high accuracy, had only 1 subject [14].

Table 6 shows that the C-SVM classifier generally gives the highest accuracy for classification of entire data set and highest accuracy overall, with 100 % accuracy for 5 subjects and 99.05 % for 3 subjects.

When comparing the results of channel reduction, it is seen that the classification using 7 channels (97.90 %) and 6 channels (98 %) has a higher accuracy than 9 studies [5–9,11,12,20,38]; using 5 channels (96.76 %) has a higher accuracy than 8 studies [6–9,11,12,20,38]; using 4 channels (95.43 %) has a higher accuracy than 7 studies [6–9,11,20,38]; and using 3 channels (92.95 %) has a higher accuracy than 5 studies [6–9,20].

Table 8 compares results with studies using the average of the accuracy from each subject. Using the Q-SVM and M-SVM classifiers in our study gives an average accuracy of 99.23 % and is higher than all other studies (Table 6).

5. Conclusion

This study investigated the effect of the number of channels and the selection of features on the classification accuracy of EMG signals acquired from the 8-channel Myo armband. The study also considered the effect of combining gyroscopic features on classification performance. The highest accuracy was 98.38 %, and was obtained using F11 + F7 + F9 as feature set using the C-SVM classifier and all channels.

Classification accuracy can be increased when the F11 feature alone is used with the gyroscopic feature; however accuracy did not improve for the highest accuracy feature set of F11 + F7 + F9.

The performance of the F11 and F11 + F7 + F9 feature sets for different numbers of channels was investigated. The addition of gyroscopic features was found to increase the classification accuracy when using a small number of channels. This would indicate that reliable results can be obtained from a reduced number of channels and using gyroscopic features. This would allow prosthetic arm control using fewer channels for hand amputees with a restricted number of muscles.

This study should be extended to amputee people to verify results and our algorithms examined for real time control of prosthetic hands.

Author statement

Both authors contributed equally to the manuscript.

Declaration of Competing Interest

The authors report no declarations of interest.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jocs.2021.101348>.

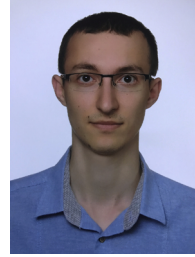
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